



A novel bronchoscopic video enhancement and tissue segmentation method based on Eulerian video magnification

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ARTICLE INFO

Keywords:

Bronchoscopic video
Eulerian video magnification
Adaptive magnification factors
Block-matching and 3D filtering
Vascular segmentation

ABSTRACT

A crucial tool for the early diagnosis of serious respiratory illness problems including lung cancer is bronchoscopy. However, the bronchoscopic video acquisition environment is constrained and unevenly illuminated, making it difficult for doctors to precisely identify intact human blood vessels and other tissue features with the naked eye. It has been demonstrated that using Eulerian video magnification techniques significantly enhances subtle changes in the video that are hardly perceptible to the human eye. We propose a novel Eulerian video linear enhancement framework for bronchoscopic video enhancement. Bronchoscopy videos are recorded and processed using Laplacian pyramid decomposition and temporal filtering algorithms to obtain the frequency bands of interest, the magnification factors of the respective bands are obtained by the constructed gain curve equations, and the magnified sub-bands are subjected to correlated noise denoising by block-matching and 3D filtering methods with exact transform-domain variance, finally we reconstruct the video. A novel framework for vessel recognition segmentation of bronchoscopic videos is proposed, thus realizing an end-to-end processing program to meet the needs of respiratory physicians. The proposed method enables remarkable enhance tissue features in bronchoscopic video and has a good effect of suppressing noise and artifacts generated during the enhancement. Tiny vessels that were previously challenging to see using the naked eye could currently be observed, and the pulsing signal of blood vessels is significantly enhanced. The experimental results demonstrate that our method is superior to the earlier method in terms of visual quality, quantitative evaluation and subjective evaluation.

1. Introduction

The health of people is seriously threatened by some respiratory-related illnesses including early invasive lung cancer and carcinoma in situ [1,2]. Patients with early-stage cancers have a strong chance of recovery, with a 5-year survival rate of 73% for stage Ia patients. However the 5-year survival rate for patients with advanced stages (stages II–IV) ranges from 9% to 46% [3].

Currently, only 16% of cancers are diagnosed at disease limitation and even fewer are diagnosed at stage 0, which results in an overall 5-year survival rate of only 15%. Early diagnosis and treatment are essential in order to improve the 5-year survival rate and prognosis of patients with the severe respiratory disease [4].

Bronchoscopy remains the most direct and effective method of detecting diseases like lung cancer. After a suspicious lung lesion has been detected by computed tomography, a decisive diagnosis can only be obtained by taking a biopsy. Cancer initiates when a lesion develops in the bronchial epithelium of the lung mucosa (the thin wall of the airways). This bronchial lesion could ultimately progress into squamous

cell carcinoma and signal the emergence of further lung tumors [5]. In order to prevent and treat disease through early identification, doctors could employ bronchoscopy to search for such lesions in the airways [6,7].

During image acquisition for tracheoscopy, patient movement (breathing) and endoscope movement can create blurring effects in some frames. In addition, there is no ambient light in the lumen other than the endoscope's on-board light source, a dynamic illumination condition that produces dark areas (in some cases up to 40% of the spatial image resolution) and different artifacts, resulting in vascular or mucosal abnormalities that are not visible even with high-definition endoscopy [8]. Therefore, enhanced endoscopic image quality is necessary to decrease the failure rate of detection of mucosal or vascular abnormalities [1,9], reduce the rate of missed clinical diagnoses, improve the efficiency and accuracy of surgeons, and improve the output of some subsequent post-processing tasks, such as 3D organ reconstruction and alignment [10,11].

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<https://doi.org/10.1016/j.bspc.2024.106042>

Received 27 June 2023; Received in revised form 20 November 2023; Accepted 29 January 2024

Available online 3 February 2024

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1.1. Related works

Certain mucosal or vascular aspects in endoscopy can be enhanced by both in-chip and post-processing technologies. Among the on-chip techniques, narrow band imaging (NBI) [12] and autofluorescence imaging (AFI) [8] are worth mentioning. Two types of NBI systems are described: A restricted spectrum of red, green, and blue light with corresponding wavelengths of 415 nm, 445 nm, and 500 nm is used for tissue illumination in an RGB sequential lighting system [13]. In another type of NBI system, NBI images are produced using bandpass filters with a bandwidth of 30 nm and centers at 415 nm (blue) and 540 nm (green) [12]. On the other hand, AFI systems use a unique rotating filter wheel in front of a xenon light source to create blue (390–470 nm) and green (540–560 nm) light for tissue illumination in that order [8]. These narrow bandwidths enhance vascular vision by reducing light scattering (the stronger the scattering, the deeper the light penetrates the mucosa).

As a consequence, these methods enhance mucosal surface contrast, shorten examination times, and do away with pointless biopsies [14]. However, all of these methods eventually make the endoscopic system's hardware more sophisticated, consume more power, while being inconvenient in clinical settings [3].

More recently, it has been meaningful and effective to use video magnification as a post-processing enhancement method for endoscopic images [15,16] which enables endoscopic image enhancement from the perspective of video sequence processing, for instance, magnifying subtle pulsations in the video, enhancing the contrast with the background and eventually identifying the vascular features in the pulsations [17].

With no additional hardware configuration required, this method instantly enhances the details of tissue features in the endoscopic video, enabling surgeons to efficiently and readily find tissues like blood veins with the naked eye [18].

The Eulerian perspective has now become the predominant perspective for video magnification, consisting of four primary steps—spatial decomposition, temporal filtering, micro signal magnification, and video reconstruction. The Laplacian pyramid is used in the Eulerian video linear magnification (EVLm) approach [19] to decompose video frames and directly magnify intensity-based motion. Although the EVLm method is computationally effective and has notable magnification effects, excessive magnification tends to introduce a lot of noise.

In contrast, the Eulerian video phase magnification (EVPm) method obtains phase-based motion [20] and speeds up the computation by decomposing the video using Riesz pyramid [21]. Deep learning-based Euler video magnification methods use convolutional neural networks to decompose video frames to learn motion representations, which can greatly reduce noise and ringing artifacts [22]. Video motion magnification methods combining hybrid temporal filtering with spatial decomposition of deep learning could precisely locate blood vessels and neurovascular bundles in robotic surgery, but it needs to be trained by natural video sequences of different complex surgical video images, and the algorithm is computationally intensive [18].

EVLm is currently used in a number of medical applications, including remote heart rate detection [23], wrist pulse frequency non-contact extraction [24], and finger vein pulsation-based biometric recognition [25]. While EVPm only enhance small motion signals in the video, EVLm is able to do the same for small color changes as well as motion signals.

In a variety of disciplines, including computer vision [26] and medical image processing [27], picture denoising has emerged as one of the most efficient methods. Spatial domain denoising and transform domain denoising are the two primary categories of noise reduction algorithms. While transforming domain-based methods operate on the sparse representation of the image in the transform domain and manipulate the transformed coefficients to minimize the noise, spatial domain schemes directly apply denoising algorithms on the image's

original pixels [28–31]. The block matching and 3D filtering (BM3D) method [32] is one of the many methods suggested in the literature, which successfully combines nonlocal and transforms domain filtering and displays good denoising performance, notably for additive Gaussian white noise. It is still regarded as among the most cutting-edge denoising techniques [33,34].

Correlation noise is widespread in many medical scene images. In x-ray medical imaging, it is typical to assume that the noise impacting the projection is smoothly correlated [35]. Thermal imaging is also affected by significant components of spatially correlated noise and fixed-mode noise, which can be considered as smoothly correlated in time [36,37]. A collaborative filtering technique has recently been developed that significantly outperforms the traditional BM3D method by precisely calculating the variance of the transform-domain correlated noise [38].

Recently, convolutional neural network (CNN) approaches have been widely popular [39,40], the success of CNNs in image denoising is attributed to their modeling capabilities and advances in network training and design. The learned models are typically tailored to particular noise levels, and deep neural networks are still ordering of magnitude more computationally complex than collaborative filters, have higher power consumption, and are less migratable [33,41]. As a consequence, the flexibility of the existing discriminative denoising methods is limited.

1.2. Contributions

In our research, we propose a novel standard white-light bronchoscopic video enhancement method that enhances the contrast of bronchoscopy images, reduces environmental noise interference. It makes doctors more easily visualize tissue abnormalities including blood vessels, including tiny blood vessels that are invisible to the naked eye. It does not require additional hardware configuration as well as a model training process. To the best of our knowledge, this is the first instance of bronchoscopy enhancement that has been done from the perspective of video processing.

We specifically provide a method for adaptive video enhancement. Each frame of a standard white light bronchoscopy video is subjected to a Laplacian pyramid decomposition to produce various spatial sub-bands, which are then banded by time-domain band-pass filtering to produce various frequency bands, and the specific magnification factor of each band is obtained by our constructed adaptive function to magnify pulsations of the vessels. Then, employing an exact calculation of the correlated noise power spectrum density to reduce the correlated noise in the image, we present an optimized multi-stage BM3D method for denoising each sub-band image. We perform video reconstruction as a final step.

We also propose a strategy for blood vessel extraction in bronchoscopic videos. It is implemented by considering the curvature properties of the tracheal wall vessels. Notably, both strategies provide an end-to-end process that can be easily deployed in a clinical setting.

We conducted experiments on clinically acquired bronchoscopic videos to verify that the proposed video enhancement model has good effect on improving bronchoscopic image quality, which enhances vascular tissue details while reducing the adverse effects of tissue fluid, artifacts from uneven light, etc. The proposed approach then demonstrated good subjective and objective performance in experiments on vascular tissue segmentation following bronchoscopic video enhancement.

The remainder of the paper is structured as follows. In Section 2, we present our framework and procedures, Section 3 details the experimental design and results, our study is concluded in Section 4.

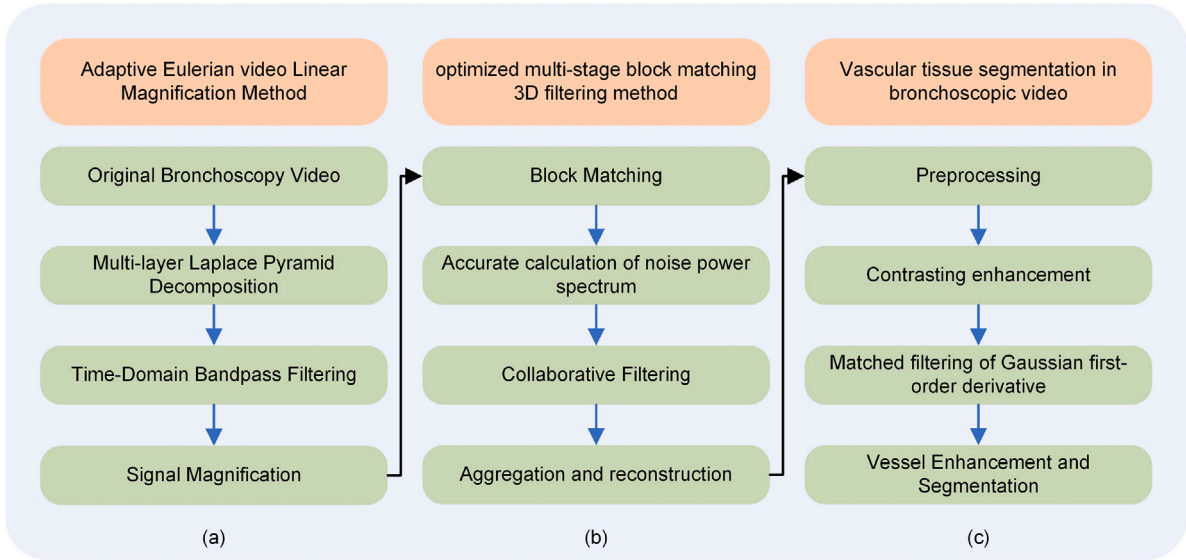


Fig. 1. A schematic diagram of our proposed framework summarizing the main operations performed in each step.

2. Methods

The conceptual diagram of our suggested framework is shown in Fig. 1. Our goal is to enhance the standard white-light bronchoscopic videos so that the vascular tissues, particularly the tiny blood vessels that are invisible to the naked eye, may be easily segmented. First, to improve the signal in the video and improve the distinction of blood vessels, mucosa, and other tissues, our video enhancement technique feeds the bronchoscopic movies into the novel adaptive Eulerian video linear magnification (AEVLM) model that has been proposed. In order to lessen the noise produced during the original video and video magnification, each spatial sub-band is denoised using an optimal multi-stage BM3D method. The processed video sequences are then assembled using Laplacian pyramid reconstruction on the video frames. Finally, we provide a method for visualization to accomplish vascular tissue segmentation in bronchoscopic videos along with a computer vision model for bronchoscopy.

2.1. Adaptive Eulerian video linear magnification method

From the viewpoint of the Eulerian perspective in fluid dynamics, EVLM is used to magnify the change in pixel values over time [19]. Small color and motion changes can both be enhanced through a joint space-time processing approach. The following analysis uses the first-order Taylor series expansion method to achieve 1D signal magnification of translational motion, which can be extended to local translational motion of 2D signals.

To briefly describe EVLM, a simple case of 1D translational motion signal $L(m, t)$ at position m and time t is taken as the analytical example. A displacement function $W(t)$ is used to express the translational motion, as described in Eq. (1):

$$\begin{cases} L(m, t) = f(m + W(t)) & t > 0 \\ L(m, 0) = f(m) & t = 0. \end{cases} \quad (1)$$

By applying the first-order Taylor approximation, the image intensity in time t can be written as:

$$L(m, t) \approx f(m) + W(t)f'(m). \quad (2)$$

Before detail magnification, the signal at the frequency of interest is filtered in time and the bandpass signal is labeled as $\psi(m, t)$. If the displacement signal $W(t)$ to be magnified is in the band of the bandpass filter, then we have:

$$\psi(m, t) = W(t)f'(m). \quad (3)$$

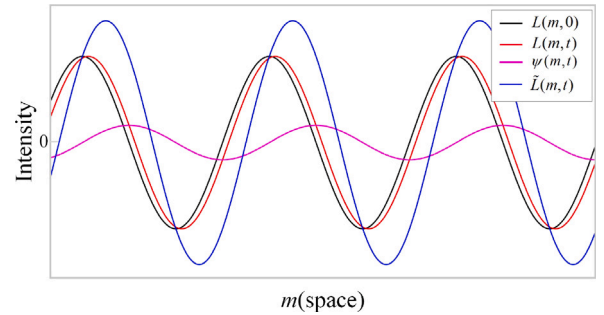


Fig. 2. Mathematical representation of EVLM for a single sinusoid.

In EVLM, the signal of the bandpass $\psi(m, t)$ is magnified by α . This magnified signal is added to the original image intensity $L(m, t)$, as shown as:

$$\tilde{L}(m, t) = L(m, t) + \alpha\psi(m, t), \quad (4)$$

where $\tilde{L}(m, t)$ represents the output signal intensity after magnification.

Combining the above equations, it can be expressed as:

$$\tilde{L}(m, t) \approx f(m) + (1 + \alpha)W(t)f'(m). \quad (5)$$

Based on the first-order Taylor assumption, the magnification of the time-band signal could be correlated with the magnification results of the motion. The output signal is generated as follows:

$$\tilde{L}(m, t) \approx f(m + (1 + \alpha)W(t)). \quad (6)$$

Fig. 2 describes the principle of motion magnification using Eulerian approach for the case of a single sinusoid.

In order to reveal subtle signals in the bronchoscopic video, our proposed video enhancement method combines spatial and temporal methods of processing. The Laplacian pyramid is used to decompose each frame of the video sequence into different spatial bands. Then, we perform band-pass filtering for each spatial band and select two first-order low-pass infinite impulse response (IIR) filters to construct an IIR band-pass filter to extract the frequency band of interest.

After that, we multiply the extracted frequency band sub-bands of interest by the magnification factor. In this process, we propose an adaptive method of magnification factor selection based on the gain curve equation.

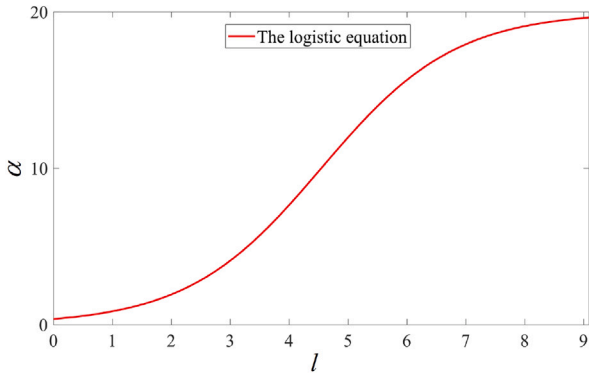


Fig. 3. Schematic diagram of adaptive selection of magnification factor.

The logistic curve equation is generally applicable to the study of the growth and development or reproductive processes of both plants and animals, in addition to social and economic events. Its main characteristics are slow growth at the beginning, rapid growth in a certain range, and slow growth after reaching a certain limit. This is a smooth “S” curve. Its basic formula is as follows:

$$S(x) = \frac{1}{1 + e^{-x}}. \quad (7)$$

The logistic equation’s curve transition is smooth, and the symmetry at both ends is about equal to the extreme value. Based on this characteristic, we develop the following gain curve equations for parameter-specific logistic equations:

$$\alpha_l = \frac{\alpha_0}{1 + \exp(\frac{-2pl}{N} + p)}, \quad (8)$$

where, α_l is the l -level magnification factor, N is the total number of pyramid levels, p is a constant, this paper takes 4.

The original spatial band is then combined with the enlarged signals to create the magnified Laplacian pyramid spatial sub-bands. Fig. 3 illustrates the adaptive selection curve of the magnification factor for a 9-level Laplacian pyramid decomposition with the magnification factor preset to 20.

2.2. Sub-bands image denoising by optimized multi-stage BM3D

After Laplacian pyramid spatial decomposition, we denoise sub-bands using an optimized BM3D algorithm. BM3D is a block matching-based image denoising algorithm that reduces noise in the 3D transform domain by measuring the similarity of blocks and by grouping sparse representations of matched blocks. There are the same three small steps in each of the two major steps of the conventional BM3D algorithm, which are separated into the basic estimate and final estimation: similar block grouping, collaborative filtering, and aggregation [32].

Similar to past efforts [42,43], we assume that medical photos contain estimated feature and noise variances. We assume that the noise in the images from bronchoscopy could be characterized as a Gaussian distribution. Using a previously established wavelet-based noise estimation method, we calculate the initial noise variance in the spatial sub-band images of bronchoscopic video frames [44], which is used as an input to an optimized BM3D method for simultaneous denoising of correlated noise at different spatial scales.

The correlation noise modeling process is as follows, let us consider a spatial sub-band image of a noisy bronchoscopic video frame $z : X \rightarrow \mathbb{R}$ of an unknown deterministic noise-free image y corrupted by stationary additive correlation noise η :

$$z(x) = y(x) + \eta(x), \quad x \in X, \quad (9)$$

where $x \in X \subset \mathbb{Z}^2$ is the coordinate in the finite 2D image domain X , and:

$$\eta = v \otimes g, \quad v(\cdot) \sim \mathcal{N}(0, 1), \quad (10)$$

with v being the zero-mean independent and identically distributed Gaussian noise with unit variance, and $\otimes$ denoting 2D convolution with the kernel g . The kernel g defines the spatial correlation of the noise as well as the noise strength, with $\|g\|_2 = \text{std}(\eta)$. An equivalent way of representing correlated noise is by its power spectral density (PSD) Ψ :

$$\Psi = E\{|\mathcal{F}[\eta]|^2\} = \text{var}\{\mathcal{F}[\eta]\} = |X| |\mathcal{F}[g]|^2, \quad (11)$$

with $\mathcal{F}$ being the 2- D Fourier transform and $|X|$ denoting the cardinality (i.e. number of elements) of X . Equivalently, a kernel g satisfying Eqs. (10), (11) can be defined from Ψ as:

$$g = |X|^{-1/2} \mathcal{F}^{-1}[\text{std}(\mathcal{F}[\eta])] = |X|^{-1/2} \mathcal{F}^{-1}[\sqrt{\Psi}]. \quad (12)$$

The process takes for input the spatial sub-band image z as well as the noisy PSD Ψ . The first step of the algorithm takes these arrays as input for hard-thresholding to produce an estimate $\hat{y}^{\text{HT}}$, then $\hat{y}^{\text{HT}}$ is processed through the global Fourier thresholding, which produces a new noisy image z^{GFTHT} and its corresponding PSD Ψ^{GFTHT} . These data are then filtered by another BM3D hard-thresholding step to obtain a new estimate $\hat{y}^{\text{HTGFT}}$. This is used as the pilot estimate for the following Wiener filter, which yields the first Wiener estimate $\hat{y}^{\text{WIE}}$. This estimate is processed again through the global Fourier thresholding filter, the output of which is a noisy image z^{GFTWIE} and its PSD Ψ^{GFTWIE} , the noise is fed to a second Wiener filter with $\hat{y}^{\text{WIE}}$ as the pilot. The output of this Wiener filter is the final estimate $\hat{y}^{\text{WIEGFT}}$.

This optimized BM3D denoising method is applied to each spatial sub-band image, and the specific implementation flow is shown in Fig. 4. This optimized BM3D denoising method is applied to each spatial sub-band image, and the specific implementation flow is shown in Fig. 5.

2.3. Vascular tissue segmentation of bronchoscopy

We propose a framework for the vascular segmentation of bronchoscopic videos, which is used to verify the efficacy of the video enhancement model proposed in this paper and to provide technical support for vascular tissue segmentation of bronchoscopic videos. Vascular tissue segmentation of bronchoscopic videos and delineation of morphological properties of vessels, such as length, width, tortuosity, branching patterns, and angles, can be used to diagnose, screen, treat, and evaluate various tracheal and pulmonary [28,30]

The input is a 3-channel RGB color bronchoscopic video frame. The basic method entails Image Preprocessing, Vessel Enhancement and Segmentation. Following are specifics of the steps of the suggested method:

Step 1: Loading the RGB video frame and extracting the green channel image.

Step 2: Contrasting enhancement and smoothening of green channel image by Contrast Limited Adaptive Histogram Equalization (CLAHE).

Step 3: Computing the complement of the image and applying Top-hat transform to correct uneven illumination in the dark background.

Step 4: Binarizing the image and applying Otsu’s method to select the optimal thresholds by maximizing the between-class variance.

Step 5: Gaussian matched filtering of bronchoscopy vessel shape features.

To extract global features and stabilize the derived vascular structure, we use a matched filtering method [45]. The features of blood vessels in the trachea and bronchi have the following characteristics: (1) Small vessel curvature can be approximated by segmented linear segments; (2) Blood vessels have lower reflectance than other surfaces and are therefore darker relative to the background; (3) Although the vessel width decreases from the optic disc, the change in vessel diameter is gradual rather than abrupt. We use these properties as the foundation for building the Gaussian matching filter.

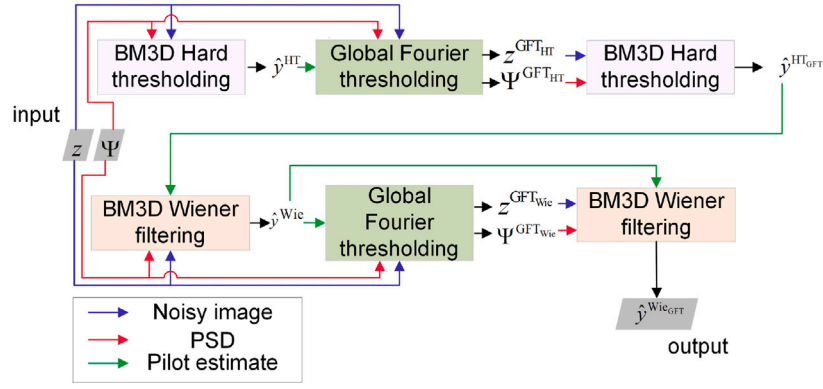


Fig. 4. Flowchart of the entire denoising process.

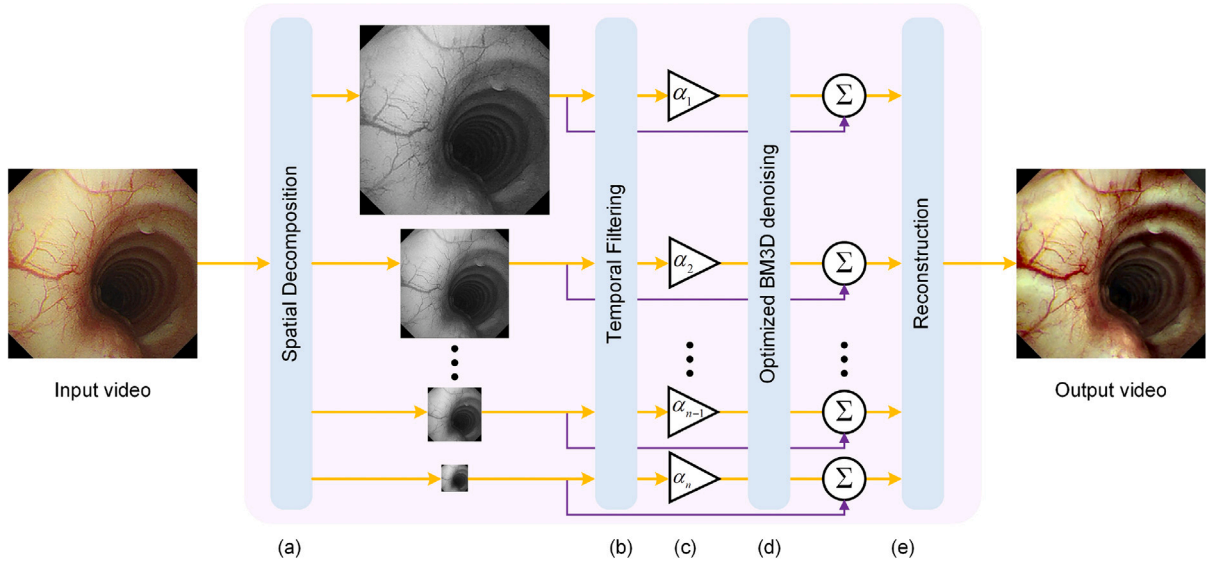


Fig. 5. Framework of the proposed video enhancement method for bronchoscopy. (a) Spatial decomposition, (b) Temporal filtering, (c) Video magnification, (d) Optimized multi-stage BM3D denoising, (e) Reconstruction magnified video frame.

Table 1
 Magnification factors of different spatial sub-bands.

Pyramid level	EVLM	AEVLM
9	20	19.6403
8	20	19.147
7	20	18.0445
6	20	15.8278
5	20	12.1864
4	20	7.8136
3	20	4.1722
2	20	1.9555
1	20	0.853

3. Experiments and results

Videos taken with the Olympus BF-Q290 bronchoscope from Qilu Hospital of Shandong University were used as experimental data, all videos have been anonymized to meet the requirements of the Institutional Review Board. On the basis of these bronchoscopic video sequences, we analyzed the proposed method in this section. On a computer with an AMD Ryzen 5 4600H CPU and 16 GB RAM, the experimental results were produced using MATLAB code. The above methods were put into practice using MATLAB 2020a. The collected video size is 1000×1000 with 25 frames per second.

The intended magnification α_0 was established by experimentation. We examined several values of α and found that a factor of 20 offers satisfactory video magnification, therefore α_0 was determined as 20. By substituting α_0 into the gain curve equation after the Laplacian pyramid decomposition of the original video, as shown in Table 1, the magnification factor α_l for each spatial sub-band was obtained. The spatial wavelength was set to 1, the chromaticity attenuation was set to 0.5, and the cutoff wavelength interval was set between 1 to 2, which corresponded to a human blood pulsation of 60 to 120 times per minute.

In our experiments, we employed the BM3D image denoising algorithm and meticulously configured its parameters to achieve optimal denoising results. The threshold for 2D block matching was set to 3, the size of 2D blocks was chosen as 8, the threshold for 3D block matching was established at 2.7, the threshold for global block matching was defined as 2500, the size of 3D blocks was specified as 8, and the Wiener filter parameter beta was set to 2.

The frequency bands of interest in each spatial sub-band are magnified and superimposed on the original spatial sub-band. Using the optimized multi-stage BM3D denoising method proposed in this paper for multi-scale denoising to reduce the magnified spatial sub-band noise. To demonstrate the superiority of our approach, we compared two video enhancement methods: EVLM and AEvLM. The bronchoscopic video enhancement effect is specifically shown in Fig. 6. Tissue features such as vessels in bronchoscopy can be visually enhanced,

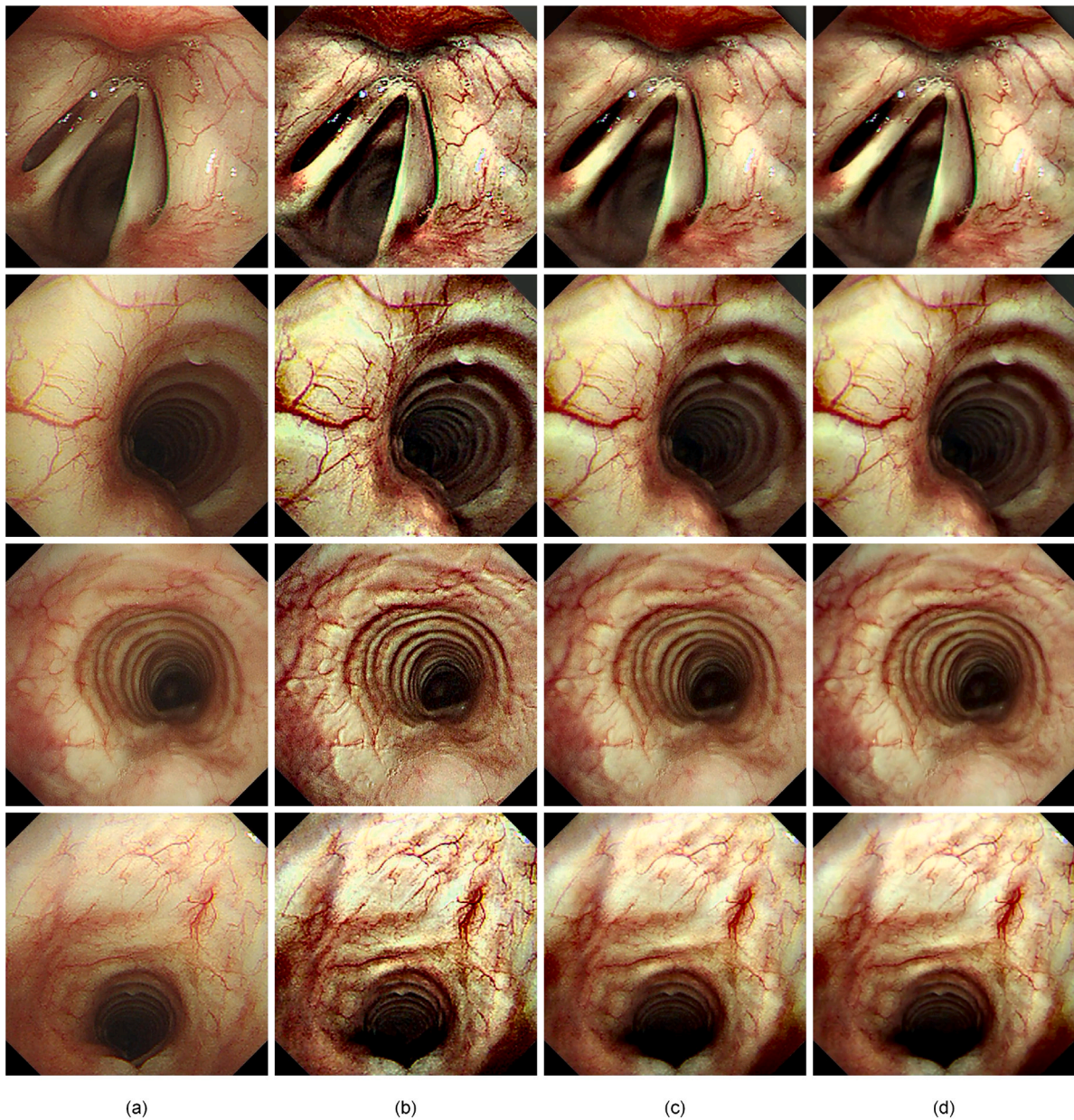


Fig. 6. The bronchoscopic video enhancement results. The first column is the original video frames. Columns (b–d) are the enhanced results of the EVLM, the AEVLM and our proposed method, respectively.

making extremely delicate vessels visible to the human eye. The EVLM method produces the most artifacts and distortions, the AEVLM method produces relatively few artifacts, but noise from the original video and video improvement is still visible, while the enhanced video with our proposed method has a clear texture of human tissue and introduces fewer noise artifacts.

We performed spatiotemporal slice control experiments to visualize motion magnification effects. Fig. 7 shows the comparison diagram of spatio-temporal slices. Select the red line position in Fig. 7(a) as the position of the slice and observe its characteristics in the spatio-temporal sequence. The results show that adaptive video magnification can retain more details in blood vessel movements, making tiny blood vessels more obvious, while greatly reducing noise in both moving areas and background areas.

Two areas of the video frame containing vascular tissue of size 400×400 were selected to create a series of 3D grayscale maps, which we call Region 1 and Region 2. The pixel grayscale information of

the original video frames and the images processed by each of the enhancement methods can be easily observed, as illustrated in Fig. 8. The intensity levels of every pixel point in the image have been enhanced to varying degrees by the video enhancement algorithm, the overall image pixel intensity after the enhancement by EVLM method is not smooth and the local variation of each pixel point was large, the pixel intensity of the enhanced image by the AEVLM method is smoother overall, but there is a relatively large magnitude of variation within the region of high-intensity pixel points, our proposed method produces image with pixel intensities is smoothest overall and smoothest within regions where each pixel intensity is similar.

The structural similarity index (SSIM) is a widely accepted and reliable method for assessing the structural similarity between two images. Table 2 shows the difference in SSIM indicators of our 3D grayscale images. The elevated SSIM scores signify a substantial improvement in the structural similarity between the original and enhanced images, indicating that our approach effectively preserves important visual details while reducing noise and artifacts. This advancement is of paramount

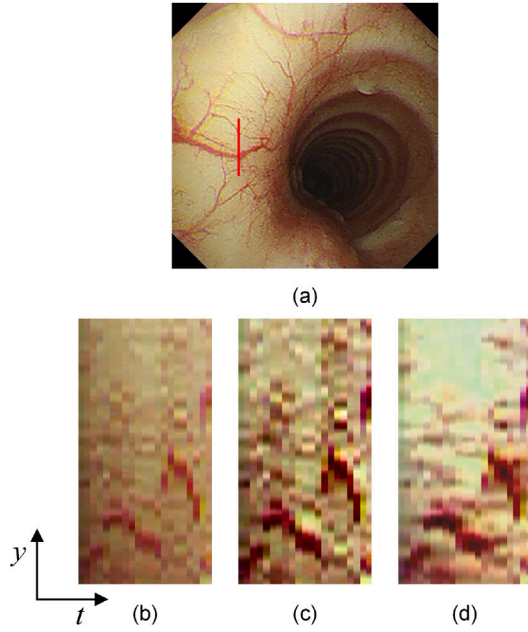


Fig. 7. Comparison of spatiotemporal slices: (a) Slice position, (b) Original, (c) EVLM, (d) AEVLM.

Table 2
SSIM comparison of 3D grayscale maps.

SSIM	EVLM	AEVLM	Ours
Region 1	0.71	0.91	0.95
Region 2	0.73	0.93	0.95

Table 3
Two commonly used image quality metrics of SSIM and PSNR for quantitative assessment of the video enhancement.

Mi(SSIM, PSNR)	EVLM	Ours
Video 1	M1(0.78, 15.05)	M2(0.86, 16.79)
Video 2	M3(0.85, 18.03)	M4(0.91, 20.72)
Video 3	M5(0.85, 17.66)	M6(0.92, 20.14)
Video 4	M7(0.92, 21.33)	M8(0.98, 26.86)

significance in the context of bronchoscopy procedures, where accurate tissue characterization and enhanced visualization of subtle structures, such as blood vessels, are pivotal for clinical diagnoses. The increased SSIM values validate the robustness of our method and its potential to provide clinicians with a more accurate and reliable tool for their medical practice.

The image preprocessing, vessel enhancement, and segmentation processes were utilized to identify and separate the vascular tissue from other tissue using Region 1 as the input. We set up two Gaussian matching filtering processes to identify thick and thin vessels, respectively, and then we combined the outcomes of the two procedures to produce the segmented vascular tissue. Fig. 9 depicts the proposed method of vascular segmentation for bronchoscopy in this paper.

Objective evaluation: In order to statistically evaluate these methods, SSIM and the peak signal-to-noise ratio (PSNR) were utilized as two objective indicators to visually compare different video enhancement techniques [46,47]. Four bronchoscopic videos were selected for processing to obtain the average of all video frame objective evaluation metrics. Calculating SSIM and PSNR between original and enhanced video for quantitative evaluation is usually reasonable as it is impossible to generate clinical ground truth data for validating endoscopic video enhancement to some extent. We also believe that SSIM can describe the consistency of human tissue pulsation between the enhanced

Table 4

Comparison of the performance of different methods for vessel segmentation.

Objective indicators	Original	EVLM	AEVLM	Ours
Accuracy	0.92	0.79	0.94	0.98
Sensitivity	0.8	0.68	0.87	0.95
Specificity	0.97	0.83	0.97	0.98
Dice	0.83	0.61	0.9	0.97

and original videos, as it primarily measures structural similarity between images, while PSNR can indicate the noise suppression effect of the video enhancement process. The results are shown in Table 3, Mi represents a set of two elements, with i indexing values from 1 to 8, corresponding to different arrays in the table. The best objective evaluation results are obtained on video 4 sequences processed by our method, with SSIM and PSNR of 0.98 and 26.86, respectively.

We compared the segmentation results with those manually annotated by experts (Ground Truth) and used different metrics to assess the performance of the method. True Positive (TP) was used to indicate how many vascular pixels were correctly segmented, False Positive (FP) to indicate how many vascular pixels are incorrectly segmented, True Negative (TN) to indicate how many background pixels are correctly segmented, and False Negative (FN) to indicate how many background pixels are incorrectly segmented.

Accuracy, Sensitivity and Specificity were used to objectively evaluate the vascular segmentation effect of the algorithm [9]. Accuracy is the ratio of correctly segmented pixels to the total pixels of the image, Sensitivity is the proportion of correctly segmented vascular pixels to the total vascular pixels, Specificity denotes the proportion of correctly segmented background pixels to the total background pixels, and The Dice similarity coefficient measures the degree of overlap between the segmentation result and the ground truth, since vascular feature segmentation often involves overlap and the accuracy of regions, the Dice similarity coefficient provides an intuitive and informative performance metric that tells us the degree of similarity between the segmentation result and ground truth. They can be obtained by the following equation:

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN}, \quad (13)$$

$$Sensitivity = \frac{TP}{TP + FN}, \quad (14)$$

$$Specificity = \frac{TN}{TN + FP}, \quad (15)$$

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN}. \quad (16)$$

Table 4 displays the objective performance evaluation indices for the various vascular methods of segmentation. With values for Accuracy, Sensitivity, and Specificity of 0.98, 0.95, and 0.98, respectively. By comparing the Dice similarity coefficient, our method has achieved a significant advantage in vascular feature segmentation, reaching the highest score of 0.97. This result demonstrates that our method excels in the degree of overlap between the segmentation results and ground truth labels. The high overlap in the segmentation results indicates that our method can more accurately capture and extract vascular features, providing medical professionals with more reliable diagnostic criteria. We also conducted the following control experiments: the original video was processed with EVLM, then subjected to our proposed BM3D filtering. The resulting values for accuracy, sensitivity, and specificity were 0.83, 0.76, and 0.85, respectively. Objective evaluation metrics were lower than those of the AEVLM method and the method proposed in this paper. The experimental results indicate that our proposed method excels in the objective assessment of tissue feature recognition capabilities.

Subjective evaluation: The respiratory surgeons manually and subjectively assessed all magnified bronchoscopic videos, and they generally

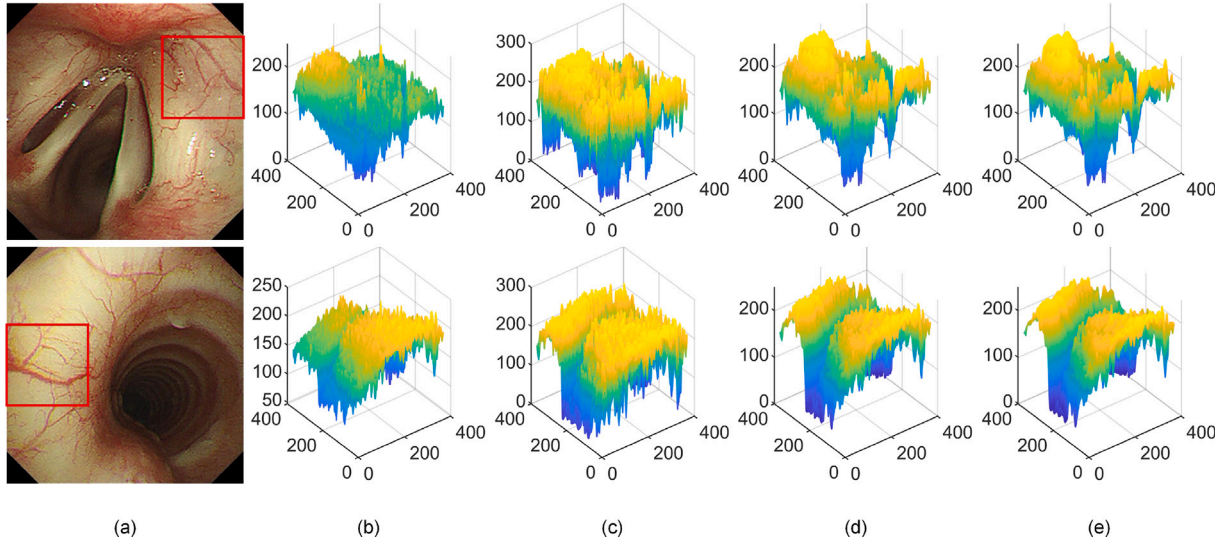


Fig. 8. 3D grayscale map. (a) Area of interest, (b) Original, (c) EVLM, (d) AEVLM, (e) Our proposed.

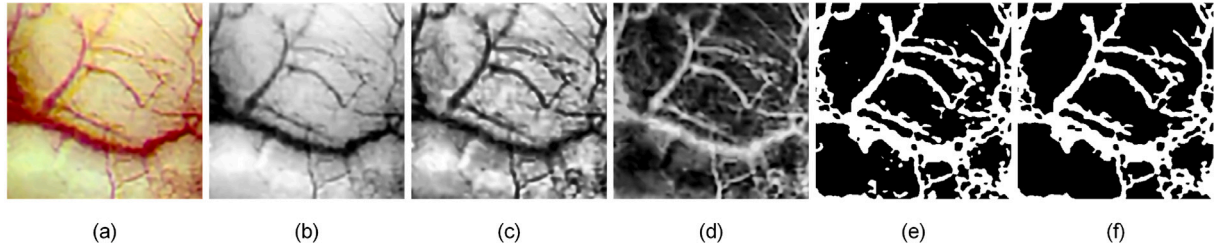


Fig. 9. Bronchoscopic vascular tissue segmentation process: (a) General image, (b) Green Channel, (c) CLAHE enhancement, (d) After top-hat, (e) Otsu thresholding, (f) Matched filtering with Gaussian derivatives.

concluded that our method could effectively help them locate human tissues in the bronchoscopic videos, such as blood vessels, and that the visual quality of the videos had better visual perception than earlier enhancement methods.

After processing the vascular tissue segmentation results with different video enhancement methods, we compared the intensity of each pixel point to the original ground truth. Different colors were used to mark them, where green represents the same intensity, red represents the pixel points that misidentified the background as blood vessels, and purple represents the pixel points of vascular tissue that are not successfully identified for segmentation. As shown in Fig. 10, EVLM method introduces significant noise, AEVLM produces less noise but identifies less complete vessels. Based on our proposed method these artifacts could be effectively eliminated while preserving the enhanced vessels, especially the vascular connections at the bifurcation are well preserved.

4. Discussion and conclusions

In clinical practice, bronchoscopy is a widely used means of effectively examining lung and tracheal lesions, and the ability to detect abnormal vascular tissue early in the video may help guide specialists to better analyze potential deep lung disease.

We realize the enhancement of bronchoscopic videos by magnifying the tiny vessel pulsation signals and changes in color intensity. Our proposed AEVLM strategy adaptively selects the appropriate magnification

factors according to the spatial frequency of each layer of the Laplacian pyramid, ensuring that the first-order Taylor series expansion could better approximate the subtle changes of the signal. The optimized multi-stage BM3D approach is used to remove correlated noise from the magnified spatial sub-bands, significantly decreasing the video noise. Respiratory and bronchoscopic motion mainly generates low-frequency noise, and both adaptive adjustment of magnification factors and spatial-domain denoising are effective in removing such noise. Our methodology strategically leverages frequency band selection, adaptive filtering techniques, and a combination of spatial and temporal noise reduction to effectively mitigate the interference caused by the motion of the bronchoscope and the patient's breathing.

The experimental results of bronchoscopic videos enhancement and vessel segmentation demonstrate that our proposed method successfully achieves video enhancement with less noise and artifacts introduced and better video quality obtained compared to the conventional Eulerian video magnification method. The best segmentation of blood vessels in bronchoscopy is achieved compared to the previous video magnification methods, resulting in clearer blood vessels. The vascular enhancement in bronchoscopic videos of the vocal cords, trachea, and bronchi could be improved using the method we propose, which requires less additional hardware configuration or model training and is more portable.

Our method still has limitations, like the fact that the magnification factor α is subjectively determined to provide good and intuitive visual

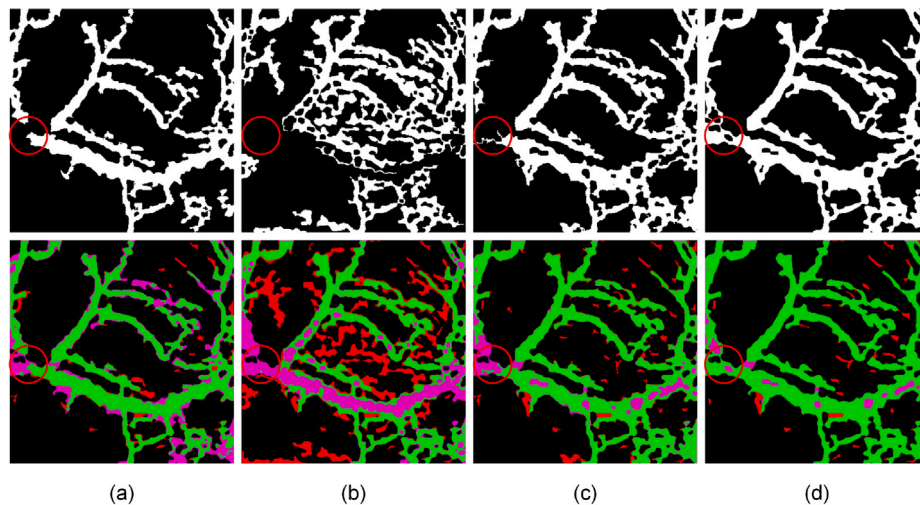


Fig. 10. Comparison of vascular tissue segmentation results and ground truth by different methods:(a) Original, (b) EVLM, (c) AEVLM, (d) Our proposed original.

quality of the experimental results. Choosing the best magnification factor for various patients is still a challenging issue in clinical practice.

In future research, we envision further exploring spatial decomposition methods in the field of video magnification. This exploration aims to tailor these methods to meet the specific requirements of various medical scenarios and applications.

To the best of our knowledge, this is the first instance of bronchoscopic video enhancement in terms of video sequence processing. We also showed that the video magnification method is workable in terms of enhancing tissue features such as endoscopic vessels. Further studies on various human endoscopic enhancement could be conducted in the future to increase clinical diagnostics and endoscopic surgery.

CRedit authorship contribution statement

Yuanzhao Yang: Conceptualization, Methodology, Validation, Writing – original draft. **Qi Jiang:** Funding acquisition, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China under Grant 61973194, in part by the Major Scientific and Technological Innovation Project of Shandong Province, China under Grant 2019JZZY010427 and in part by the Shenzhen Fundamental Research and Discipline Layout Project, China under Grant JCYJ20190806155616366. We thank Qilu Hospital of Shandong University for providing the data with ethical approval.

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